

AI, Green Technology, and Structural Change: OECD and Emerging-Market Evidence, with Focus on Vietnam and Southeast Asia

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Abstract Artificial intelligence and green technology are advancing as general-purpose technologies whose effects on development outcomes remain incompletely understood outside the advanced economies where most existing evidence is concentrated. This essay asks whether artificial-intelligence patent intensity and green technology intensity are associated with structural change along three margins of development -- carbon emission intensity, labor productivity, and the labor income share -- and whether these relationships differ between advanced-economy and emerging-market industries, with particular attention to Vietnam and Southeast Asia. It builds a unified two-way fixed-effects and shift-share instrumental-variable framework applied consistently across six country-industry panels, and decomposes the labor-share margin into its wage and employment channels. In advanced-economy industries, higher artificial-intelligence patent intensity is associated with lower carbon emission intensity, while its aggregate effect on the labor income share is negligible; decomposing that result reveals two large offsetting forces, an employment gain alongside a wage discount, so the null result masks real redistribution between headcount and wage share. Green technology intensity raises labor productivity in these economies, more so in industries with high artificial-intelligence adoption and, regionally, most strongly within Southeast Asia. In the emerging-market panel, the labor-share relationship is modestly positive and wage-driven, though weaker within Southeast Asia specifically. These findings suggest technology adoption need not come at the cost of either the environment or labor's income share, but how gains split between employment and wages is a distinct policy question, directly relevant to how Vietnam should monitor its own technology-driven structural transformation.

Keywords artificial intelligence, green technology, labor income share, carbon emission intensity, Vietnam

1 Background

1.1 Motivation and development context

Artificial intelligence and green technology are general-purpose technologies, broad and fast-improving enough to reshape production well beyond the sector where they first appear (Bresnahan and Trajtenberg, 1995; Brynjolfsson, Rock and Syverson, 2021). Acemoglu and Restrepo (2020, 2022) show that AI reorganizes tasks within production, while green technology pushes the input mix toward lower-emission techniques (Popp, 2019; Albrizio, Kozluk and Zipperer, 2017). What matters is how the two interact with production. Two industries can adopt AI at identical speed and still end up with different emissions profiles, and a gain in green productivity does not automatically split evenly between capital and labor. For that reason, this essay works with three margins of structural change - environmental, efficiency, distributional - inside a single technology-intensity framework.

Development level changes what technology adoption looks like. OECD economies sit at the mature end: patenting concentrates there, and industry-level data are complete. Emerging markets, by contrast, are latecomers in Gerschenkron's sense (1962; see also Perez and Soete, 1988) - they import frontier technology more often than they generate it. That fact bears directly on the middle-income-trap question of whether catching-up economies can convert imported technology into productivity gains without stalling in a low-value-added equilibrium (Lin, 2012; Paus, 2012).

1.2 Vietnam and Southeast Asia

Vietnam's position is a distinctive one. The country sits deep in global manufacturing value chains, hosting assembly operations for multinational firms, yet it holds only a small share of the higher value-added design stages (World Bank, 2021). At the same time, it has committed to accelerating AI adoption and reaching net-zero emissions by 2050. Those two commitments pull against each other. AI-driven automation could deliver the productivity gains needed to escape the middle-income trap, but only if it does not slow the emissions transition or pull gains away from labor. Whether that tension actually shows up in Vietnam, and across the wider Southeast Asian (SEA) group, is one of this essay's central applied contributions.

1.3 Contribution and research questions

The applied contribution of this essay runs along three lines. First, it brings carbon emission intensity, labor productivity and the labor income share into a single fixed-effects instrumental-variable framework, so the three outcomes can be compared on a like-for-like basis rather than treated separately. Second, each

technology-intensity regressor gets its own shift-share (Bartik-style) instrument, validated through the Rotemberg-weight diagnostics proposed by Goldsmith-Pinkham, Sorkin and Swift (2020). Third, the labor-share margin is decomposed into offsetting wage and employment channels and extended regionally to ASEAN - a step that shows how a null average effect can hide redistribution that a single-coefficient table would never reveal.

Three research questions organize the empirical work. RQ1: does AI patent intensity raise or lower carbon emission intensity at the country-industry level, and does this differ between OECD and emerging-market industries? RQ2: does green technology intensity raise labor productivity, and is this dividend moderated by an industry's prior AI adoption? RQ3: does AI patent intensity raise or lower the labor income share, and, if the aggregate effect is near zero, through which channel, wage or employment, is that outcome generated?

2 Theoretical Framework

2.1 A task-based framework for technology and structural change

The task-based framework of Acemoglu and Restrepo (2020, Journal of Political Economy; 2022, Econometrica) organizes all three margins. Automation reallocates tasks between capital and labor through three offsetting channels: displacement, where technology takes over labor tasks, lowering labor's value-added share and, if those tasks were energy-intensive, emission intensity too; productivity, which lowers production cost and raises labor demand in remaining tasks; and reinstatement, which creates new tasks where labor retains a comparative advantage, raising employment and its share. These channels move in different directions and at different speeds, so the framework offers no unambiguous prediction for AI's net effect on any margin. The effect is, in the end, an empirical question - and decomposing it channel by channel can tell us more than the aggregate number ever would.

2.2 How AI and green technology intensity are measured

The analysis anchors on AI Patent Intensity (APII), built by Baruffaldi et al. (2020, OECD) from EPO patent families and re-weighted onto industries through the ALP CPC-to-ISIC crosswalk. It carries the analysis throughout for a practical reason: occupation-exposure indices such as Felten, Raj and Seamans's (2021) AIOE exist almost exclusively for OECD labor markets, while APII keeps the OECD and EM panels comparable to each other. Green Technology Intensity (GTI) comes from EPO ENV-TECH Y02 patents (OECD) or the WIPO Green Inventory

(EM). The green-productivity link draws on the Porter hypothesis (Porter and van der Linde, 1995): Albrizio, Kozluk and Zipperer (2017) find environmental stringency associated with short-run productivity gains in the most advanced industries, where process innovation exceeds compliance cost for firms with absorptive capacity (Cohen and Levinthal, 1990; Popp, 2019).

2.3 The labor income share and its wage-employment decomposition

The labor-share literature has moved from documenting the aggregate decline in labor's value-added share (Karabarbounis and Neiman, 2014; Rognlie, 2015) toward explaining its industry- and firm-level sources. Autor, Dorn, Katz, Patterson and Van Reenen (2020) trace much of the decline to superstar firms: these capture a growing share of value added while paying labor a shrinking one. Labor-share comparisons are sensitive to self-employment income, which often gets misclassified as capital income - for that reason, this essay applies the Gollin (2002) adjustment as a robustness check.

2.4 Carbon, productivity and labor share in one model

These strands converge on a single hypothesis. AI patent intensity captures the carbon and labor-share margins; green technology intensity captures productivity. Both, the essay argues, are associated with structural change across the environmental, efficiency and distributional margins. Each is estimated on a mature, complete OECD panel and a catching-up EM panel where data completeness binds precision. Moderators (high-renewable, high-AI, high-capital) test whether effects strengthen where preconditions are favorable, and an ASEAN interaction in every part tests whether Vietnam's regional peers differ from the wider EM panel. That effects should be more consistent in the OECD than the still-developing EM panel is further motivated by Acemoglu (2025), Eloundou et al. (2024) and Graetz and Michaels (2018).

3 Data

3.1 Sample architecture and sources

Table 1 summarizes the sample architecture of six country-industry panels. Part A pairs 25 OECD countries with three ISIC sectors for 2010-2022 (75 pairs, 936 observations) and 43 EM countries (74 pairs, 591 observations, 2012-2022). Part B

pairs the same 25 OECD countries with a broader ENV-TECH industry set (175 pairs, 1,847 observations, 2012-2022) and an EM panel of 72 pairs (536 observations). Part C uses the ten-industry STAN A*10 classification, pairing 25 OECD countries with ten industries (250 pairs, 3,120 observations, 2010-2022) and 39 EM countries (67 pairs, 530 observations). The smaller EM cluster counts are the recurring statistical-power constraint referenced below.

Table 1. Sample Architecture and Headline FE-IV Estimates Across Parts A-C

Part	Sample	Outcome	Key regressor	Countries	Pairs	Obs	Years	beta (FE-IV)	SE	p<0.10
A	OECD	ln CO2 intensity	AIPI	25	75	936	2010-2022	-0.0965	0.0327	Yes
	EM	ln CO2 intensity	AIPI_EM	43	74	591	2012-2022	0.8012	0.6953	No
B	OECD	ln labour productivity	ln_GTI	25	175	1847	2012-2022	0.0187	0.0069	Yes
	EM	ln labour productivity	ln_GTI_EM	43	72	536	2012-2022	0.2132	0.2033	No
C	OECD	ln labour income share	AIPI	25	250	3120	2010-2022	0.0114	0.0255	No
	EM	ln labour income share	AIPI_EM	39	67	530	2012-2022	0.4550	0.3392	No

Pairs = country x industry. beta = coefficient on the part's key technology regressor (FE-IV main spec).

The OECD analysis draws on OECD STAN for value added, employment and capital; the OECD AI Patents database for AI Patent Intensity; EPO ENV-TECH Y02 patents for Green Technology Intensity; EDGAR sectoral CO2 emissions, a country-level series without industry breakdown (a limitation carried to the conclusion); and the World Bank's WDI for controls (GDP per capita, renewable share, R&D, trade openness, broadband, tertiary enrollment). The EM analysis draws on a parallel set of sources: UNIDO INDSTAT4 (ISIC sectors C and D_E), WIPO's AI patent data and Green Inventory, and the World Governance Indicators.

3.2 Vietnam and Southeast Asia in the data

Vietnam's presence in the data varies by part - and that's a coverage constraint, not a modeling choice. In Part B it enters the main EM panel directly, alongside Indonesia, Laos, Malaysia, the Philippines and Thailand. In Parts A and C, EPO AI-

patent filings from Vietnamese applicants are too sparse for an industry-level series, so Vietnam instead enters an alternate WIPO-based panel (Table 9). Mainland China and South Africa are absent from the labor-share panel because UNIDO INDSTAT4 does not report the value-added series needed.

3.3 Descriptive statistics and trends

Table 2 reports pooled descriptive statistics for the three OECD panels. AI Patent Intensity averages 1.41 (SD 1.83) in the three-sector Part A panel and 0.91 (SD 1.29) in the ten-industry Part C panel, reflecting different classification denominators. Green Technology Intensity is highly right-skewed, averaging 4,608 green patents per billion USD of value added (SD 13,388). CO2 emission intensity (in logs) averages 5.86 (SD 2.22), labor productivity 5.58 (SD 1.95), and the labor income share -0.84 (SD 0.45), a raw labor share of roughly 43 percent, consistent with published OECD series.

Table 2. Descriptive Statistics: Key Variables, Parts A-C (OECD Panel)

Variable	N	Mean	SD	Min	Max
Part A - AI Patent Intensity and Carbon Intensity (OECD)					
AI Patent Intensity (ln, industry-weighted AI patents)	975	1.4100	1.8300	0.0000	7.9700
CO2 emission intensity (ln, tonnes CO2 / mn USD VA)	975	5.8600	2.2200	-1.6400	9.0900
Part B - Green Technology Intensity and Labor Productivity (OECD)					
Green tech. intensity (Y02 patents per bn USD VA)	2274	4607.6400	13388.4700	0.0000	146665.7600
Labor productivity (ln, thousand USD VA per worker)	2274	5.5800	1.9500	2.9200	12.4200
Part C - AI Patent Intensity and Labor Income Share (OECD)					
AI Patent Intensity (ln, industry-weighted AI patents)	3250	0.9100	1.2900	0.0000	7.7500
Labor income share (ln, LABR / VALU)	3250	-0.8400	0.4500	-3.1300	0.4800

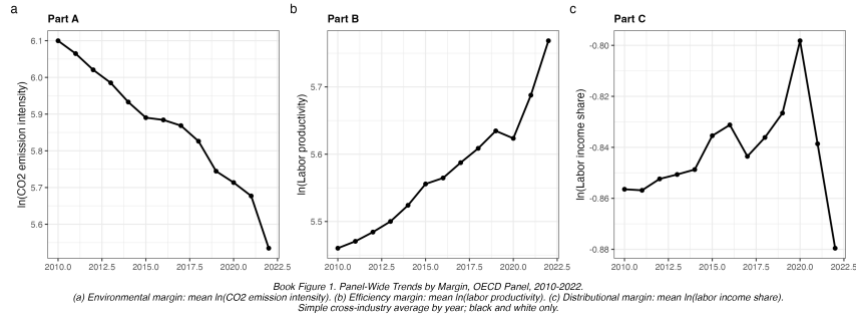


Fig. 1. Panel-Wide Trends by Margin, OECD Panel, 2010-2022. (a) Environmental margin: mean ln(CO2 emission intensity). (b) Efficiency margin: mean ln(labor productivity). (c) Distributional margin: mean ln(labor income share).

Figure 1 plots the pooled OECD trend of all three outcomes over 2010-2022. Carbon intensity trends gently downward, labor productivity trends upward, and the labor income share is essentially flat across the period, a pattern that foreshadows the offsetting-channels result in the results section below.

4 Model

Each part shares a common two-way fixed-effects and shift-share instrumental-variable structure, but with its own outcome, key regressor and control set. Let i index a country-industry pair and t index year. **Part A (environmental margin)** models log carbon emission intensity on AI patent intensity (AIPi), Equation 1:

$$\ln(CO2Int_{it}) = \alpha_i + \lambda_t + \beta_1 \cdot AIPi_{it} + \beta_3 \cdot \ln(GDPpc_{it}) + \beta_4 \cdot Renew_{it} + \beta_5 \cdot CapForm_{it} + \beta_6 \cdot RD_{it} + \beta_7 \cdot Trade_{it} + \beta_8 \cdot Broadband_{it} + \varepsilon_{it} \quad (1)$$

Part B (efficiency margin) replaces the outcome with log labor productivity and the key regressor with log green technology intensity (GTI), and uses tertiary school enrollment in place of trade openness, Equation 2:

$$\ln(prod_{it}) = \alpha_i + \lambda_t + \beta_1 \cdot \ln(GTI_{it} + 1) + \beta_3 \cdot \ln(GDPpc_{it}) + \beta_4 \cdot Renew_{it} + \beta_5 \cdot CapForm_{it} + \beta_6 \cdot RD_{it} + \beta_7 \cdot Tertiary_{it} + \beta_8 \cdot Broadband_{it} + \varepsilon_{it} \quad (2)$$

Part C (distributional margin) models the log labor income share, again on AI patent intensity, Equation 3:

$$\ln(labshare_{it}) = \alpha_i + \lambda_t + \beta_1 \cdot AIPi_{it} + \beta_3 \cdot \ln(GDPpc_{it}) + \beta_4 \cdot Renew_{it} + \beta_5 \cdot CapForm_{it} + \beta_6 \cdot RD_{it} + \beta_7 \cdot Trade_{it} + \beta_8 \cdot Broadband_{it} + \varepsilon_{it} \quad (3)$$

In every part α_i is a country-industry pair fixed effect absorbing time-invariant heterogeneity, λ_t a year fixed effect absorbing common shocks, and ε_{it} an

idiosyncratic error. Moderated specifications add an interaction term, where HighX is HighRenewable (Part A), HighAI (Part B) or HighCapital (Part C), Equation 4:

$$y_{it} = \alpha_i + \lambda_t + \beta_1 \cdot Tech_{it} + \beta_2 \cdot Tech_{it} \times HighX_i + \beta_3 \cdot \ln(GDPpc_{it}) + \beta_4 \cdot Renew_{it} + \beta_5 \cdot CapForm_{it} + \beta_6 \cdot RD_{it} + \beta_7 \cdot Trade_{it} + \beta_8 \cdot Broadband_{it} + \varepsilon_{it} \quad (4)$$

Because the key technology regressor is potentially endogenous, an industry may patent more because it anticipates the very outcome under study, each part instruments it with a leave-one-out shift-share (Bartik) instrument. Equation 5 is the first stage for Parts A and C, which share the same AIPI regressor and its AIPI_bartik instrument:

$$AIPI_{it} = \pi_1 \cdot AIPI_bartik_{it} + \pi_2 \cdot \ln(GDPpc_{it}) + \pi_3 \cdot Renew_{it} + \pi_4 \cdot CapForm_{it} + \pi_5 \cdot RD_{it} + \pi_6 \cdot Trade_{it} + \pi_7 \cdot Broadband_{it} + \alpha_i + \lambda_t + u_{it} \quad (5)$$

The **Part B first stage** is identical in form, with log green technology intensity instrumented by GTI_bartik and tertiary enrollment among the controls. The instrument (AIPI_bartik_{it} or GTI_bartik_{it}) is the leave-one-out mean of the technology measure across the other 24 OECD countries in the same year, interacted with each industry's initial technology composition; the estimation and testing section below reports the diagnostics supporting its relevance and exogeneity.

For **Part C**, since the labor income share is by construction the wage bill divided by value added, and value added is the product of employment and average output per worker, the log labor share obeys the exact accounting identity, Equation 6:

$$\ln(labshare_{it}) = \ln(avgwage_{it}) + \ln(employment_{it}) - \ln(valueadded_{it}) \quad (6)$$

Part C exploits this identity by re-estimating the second stage of Equation 3 separately with average wage, employment and value added as the dependent variable, verifying that the three channel coefficients sum back to the directly estimated labor-share coefficient. **That reversal - two large, significant effects pulling in opposite directions once the null aggregate is decomposed - is the central empirical finding of this essay.**

The estimation climbs a ladder of four methods. Pooled OLS comes first, purely as a naive comparator. Two-way fixed effects follows, then fixed-effects instrumental variables - the preferred specification, given the endogeneity tests discussed below. A dynamic panel GMM estimator closes the ladder, added as a further robustness check on the labor-share margin.

5 Estimation and Testing

Every FE-IV specification is accompanied by diagnostic tests reported in Table 3 for all three OECD panels at alpha = 0.10: multicollinearity (max VIF), Hausman (FE vs RE), Wooldridge serial correlation, first-stage F, Wu-Hausman endogeneity,

and Arellano-Bond AR(1)/AR(2) for the dynamic GMM specification. An eighth test, Hansen's J overidentification, is the primary exogeneity diagnostic, reported as instruments "may not be exogenous" if rejected rather than claiming positive proof if not.

Table 3. Diagnostic Tests, Parts A-C (OECD Panel, $\alpha = 0.10$)

Test	Statistic (A / B / C)	p-value (A / B / C)	Decision
Multicollinearity (max VIF)	Max VIF: 1.6200 1.6400 1.6500		No severe multicollinearity in any part
Hausman FE vs RE	chi2: 3.1270 63.5130 252.8340	1.0000 0.0000 0.0000	FE retained in all parts (theory; RE rejected in B, C)
Serial correlation (Wooldridge)	chi2: 350.0400 786.2140 1184.2380	0.0000 0.0000 0.0000	Present in all parts; cluster-robust SE applied
Weak instrument (first-stage F)	F: 2321.7700 43949.6100 4906.2300	0.0000 0.0000 0.0000	Relevant instrument ($F > 10$) in all parts
Endogeneity (Wu-Hausman)	F: 21.4000 12.8800 63.2100	0.0000 0.0003 0.0000	Endogenous in all parts; IV justified
AR(1) – Arellano-Bond	z: -0.6710 -1.6030 -3.7740	0.5030 0.1090 0.0002	Fail to reject in A, B (inconclusive, low power); rejected in C (AR(1) present, expected)
AR(2) - Arellano-Bond	z: -1.4620 -0.4800 -1.5700	0.1440 0.6310 0.1160	Valid instruments (fail to reject H_0) in all parts
Hansen J (overidentification)	chi2: 3.0910 2.4470 3.6360	0.3780 0.4850 0.3040	Instruments may be exogenous (fail to reject H_0) in all parts

First-stage relevance is satisfied in every part ($F = 2,321.8$ Part A, 43,949.6 Part B, 4,906.2 Part C, all far above 10; Table 7 reports the Part C first-stage coefficient of 0.7316, significant at 1 percent). Wu-Hausman rejects exogeneity in every part ($F = 21.4, 12.9, 63.2$; all $p < 0.001$), so instrumenting is warranted. Hausman rejects random effects in Part B ($\chi^2 = 63.5$) and Part C ($\chi^2 = 252.8$) but not Part A ($\chi^2 = 3.1$); fixed effects are retained on theoretical grounds. Hansen's J fails to reject overidentification in every part ($\chi^2 = 3.1/2.4/3.6$, $p = 0.378/0.485/0.304$). Serial correlation is present (Wooldridge), so standard errors are clustered at the pair level throughout.

Because the EM panels have far fewer clusters (67 to 75 pairs against 75 to 250), cluster-robust inference can be unreliable, so a wild cluster bootstrap is applied as a fallback, an inherited limitation. Exogeneity is further probed by a Goldsmith-Pinkham, Sorkin and Swift (2020) Rotemberg-weight decomposition, which shows the carbon and labor-share margins draw most identifying variation from manufacturing.

As a descriptive complement, never a substitute for the causal estimates, TreeSHAP variable-importance analysis (Lundberg and Lee, 2017) shows country-level controls carry as much average importance as the technology regressor itself, with a near-monotonic dependence pattern consistent with the linear-in-logs form used throughout.

6 Result and Analysis

6.1 The environmental and efficiency margins

Table 4 reports the full FE-IV specification for all three parts side by side. In the OECD panel, the FE-IV estimate of AI Patent Intensity on log CO2 emission intensity is -0.0965 (SE 0.0327, significant at 1 percent): higher AI patent intensity is associated with lower carbon emission intensity, consistent with a technique effect dominating any scale effect at the OECD frontier. The emerging-market panel returns a positive coefficient (0.8012, SE 0.6953), though it can't be statistically distinguished from zero. With only 74 pairs and a short window to work with, that imprecision is best read as a limitation of the data, not as evidence that AI adoption raises EM carbon intensity.

Table 4. FE-IV Estimates: Environment, Efficiency and Distribution, OECD vs EM

	Part A OECD: CO2 int.	Part A EM: CO2 int.	Part B OECD: Product.	Part B EM: Product.	Part C OECD: Lab. share	Part C EM: Lab. share
AI Patent Intensity (ln, IV) [OECD]	-0.0965***				0.0114	
	(0.0327)				(0.0255)	
AI Patent Intensity (ln, IV) [EM]		0.8012				0.4550
		(0.6953)				(0.3392)

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	Part A OECD: CO2 int.	Part A EM: CO2 int.	Part B OECD: Product.	Part B EM: Product.	Part C OECD: Lab. share	Part C EM: Lab. share
Green Tech Intensity (ln, IV) [OECD]			0.0187***			
			(0.0069)			
Green Tech Intensity (ln, IV) [EM]				0.2132		
				(0.2033)		
Num.Obs.	936	591	1847	536	3120	530
R2	0.9970	0.9370	0.9980	0.9350	0.9350	0.8490
R2 Within	0.1680	0.0410	0.1430	0.1220	0.1520	0.1130
FE: year	Yes	Yes	Yes	Yes	Yes	Yes
FE: pair_id_em		Yes		Yes		Yes
FE: pair_id	Yes		Yes		Yes	

p<0.10, ** p<0.05, *** p<0.01.

Each column re-estimates the main FE-IV specification for its part: Part A, AIPI → ln CO2 emission intensity; Part B, ln GTI → ln labor productivity; Part C, AIPI → ln labor income share. Samples and instruments differ by design across parts. SE clustered at pair level. Country-level controls (GDP per capita, renewable energy, capital formation, R&D, trade openness, broadband) included in every specification; coefficients omitted for space.

Green Technology Intensity lifts OECD labor productivity, and not by a trivial margin: the FE-IV estimate lands at 0.0187 (SE 0.0069, 1 percent), fitting the Porter-hypothesis logic outlined earlier. Switch to two-way fixed effects, and above-median AI exposure roughly triples that dividend (0.0420, 5 percent, not tabulated) - likely complementarity at work, since strong AI-driven process capability helps a green patent turn into an actual shop-floor gain rather than staying on paper. The EM panel is murkier alone: positive (0.2132, SE 0.2033) but short of significance. ASEAN breaks the pattern, though, with a significantly larger dividend than the rest of the EM group (interaction 0.198, 10 percent; Table 8) - arguably the essay's strongest positive regional signal.

6.2 What AI patent intensity does to labor share

Table 5 holds the main result of this essay - the labor-share margin. In the OECD panel, the FE-IV estimate of AI Patent Intensity on the log labor income share is 0.0114 (SE 0.0255), a coefficient that can't be told apart from zero. Taken by itself, that null tells us almost nothing: it could mean no effect, an effect too small to pick up, or two effects cancelling each other out.

Table 5. Main Results: AI Patent Intensity and the Labor Income Share (OECD Panel)

	Pooled OLS	FE	FE+Interaction	FE-IV	Diff. GMM
AI Patent Intensity (ln)	0.0635*** (0.0186)	-0.0163 (0.0160)	-0.0110 (0.0153)		-0.0020 (0.0192)
AI Patent Intensity (ln, IV)				0.0114 (0.0255)	
AI Patent Intensity x High-capital dummy			-0.0058 (0.0052)		
Lagged labor share (t-1)					0.6509*** (0.1032)
Num.Obs.	3120	3120	3120	3120	
R2	0.1100	0.9350	0.9350	0.9350	
R2 Within		0.1530	0.1540	0.1520	
FE: pair_id		Yes	Yes	Yes	
FE: year		Yes	Yes	Yes	

p <0.10, ** p<0.05, *** p<0.01.

Dependent variable: $\ln(\text{labor income share}) = \ln(\text{LABR} / \text{VALU})$. FE-IV instrument: AIPI_bartik (leave-one-out mean AI patents across the other 24 OECD countries, same year). SE clustered at pair level (Cols 0-3); two-step robust GMM SE (Col 4). Country-level controls (GDP per capita, renewable energy, capital formation, R&D, trade openness, broadband) included in every specification; coefficients omitted for space.

The ambiguity does not survive Table 6. Swapping the aggregate for its three components - employment, average wage, value added - and re-running the FE-IV specification, a clearer picture emerges: employment climbs by 0.0899 (SE 0.0226, 1 percent), value added rises but more noisily (0.0079, SE 0.0321), and the average wage falls by 0.0706 (SE 0.0226, 1 percent) - a drop large enough to matter on its own. None of the three stands alone. Equation 6 ties them together by construction: 0.0899 minus 0.0706 minus 0.0079 lands exactly on 0.0114, the original aggregate, with nothing left over (Figure 2a). Put together, the story is specific: AI patent intensity puts more people to work, but at a wage discount steep enough to erase almost all of the gain - employment grows, labor's wage bill doesn't. That is arguably the paper's most important result, one a single aggregate coefficient would have hidden completely.

Table 6. Channel Decomposition: Employment, Wage and Value-Added Effects (FE-IV)

	Employment	Average wage	Value added
AI Patent Intensity (ln, IV)	0.0899*** (0.0226)	-0.0706*** (0.0226)	0.0079 (0.0321)
Num.Obs.	3120	3120	3120
R2	0.9990	0.9990	0.9980
R2 Within	0.1060	0.2050	0.2210
FE: pair_id	Yes	Yes	Yes
FE: year	Yes	Yes	Yes

p<0.10, ** p<0.05, *** p<0.01.

Dependent variables: (a) ln(Total employment); (b) ln(Average wage); (c) ln(Value added). Same FE-IV specification and instrument (AIPi bartik) as Book Table 3, col (3). Identity: $\ln(\text{labor share}) = \ln(\text{avg. wage}) + \ln(\text{employment}) - \ln(\text{value added})$. SE clustered at pair level. Country-level controls (GDP per capita, renewable energy, capital formation, R&D, trade openness, broadband) included in every specification; coefficients omitted for space.

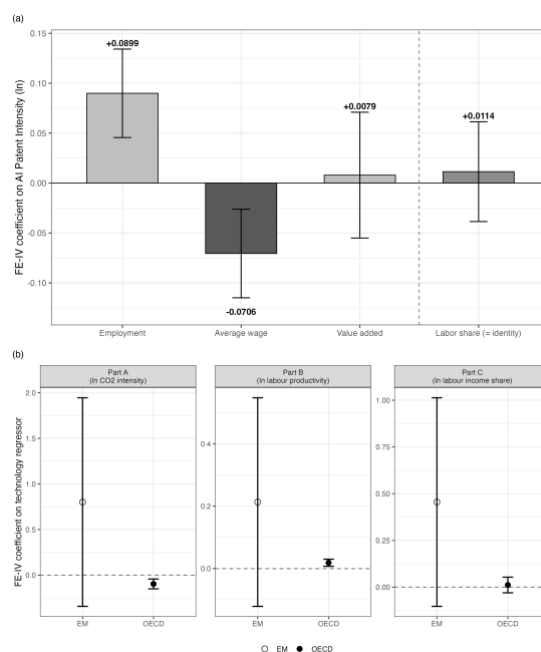


Fig. 2. Fig. 2. Results overview, Part C and cross-part (FE-IV). (a) Channel decomposition, Part C: wage + employment - value added = labor share (identity gap = 0.0000). (b) Headline FE-IV estimates, Parts A-C, OECD (filled circle) vs EM (open circle); error bars = 90% CI, SE clustered at pair level.

In the EM panel, FE is preferred over FE-IV here - the Wu-Hausman test doesn't reject exogeneity - and it gives a positive, marginally significant coefficient of 0.0622 (SE 0.035, 10 percent). The channel mix differs from OECD: wage is positive (0.047, not significant), employment flat (0.001), value added small and negative (-0.015). None is individually significant, but wage carries by far the largest estimate - fitting for industries earlier in the adoption stage. Within ASEAN, that positive EM relationship weakens or reverses (interaction -0.102, significant at 5%, Table 8).

Table 7 backs the identification strategy too: splitting by country-median capital intensity shows no significant OECD gap between high-capital (-0.0133) and low-capital (0.0365) industries - so the OECD null isn't just an artifact of pooling dissimilar industries together.

Table 7. Instrument Validity and Subsample Robustness (OECD Panel)

	First stage (AIPi ~ AIPi_bartik)	(R1a) High-capital	(R1b) Low-capital
AIPi_bartik [instrument]	0.7316*** (0.0402)		
AI Patent Intensity (ln, IV)		-0.0133 (0.0341)	0.0365 (0.0336)
Num.Obs.	3120	1549	1553
R2	0.9660	0.9270	0.9410
R2 Within	0.3970	0.1840	0.0540
FE: pair_id	Yes	Yes	Yes
FE: year	Yes	Yes	Yes

p < 0.1, ** p < 0.05, *** p < 0.01

First stage: DV = AIPi; instrument = AIPi_bartik (Bartik shift-share, other 24 OECD countries). R1: FE-IV on high-capital (HighCapital=1) vs low-capital (HighCapital=0) subsamples, country-specific median threshold. SE clustered at pair level. Country-level controls (GDP per capita, renewable energy, capital formation, R&D, trade openness, broadband) included in every specification; coefficients omitted for space.

6.3 Vietnam and Southeast Asia in focus

Table 8 brings together the three ASEAN-specific interaction results, estimated by adding a $Tech_{it} \times ASEAN_i$ interaction to each part's EM specification (Equations 1-3). On carbon intensity, no detectable ASEAN-specific deviation emerges (interaction 0.019, SE 0.133, not significant). On productivity, the essay finds its clearest positive regional result: the green-technology productivity dividend is significantly larger in ASEAN than the rest of the EM panel (0.198, SE 0.110, significant at 10 percent). On the labor share it finds the opposite: the modest

positive AI-labor-share relationship in the wider EM panel is significantly weaker, or reversed, within ASEAN (-0.102, SE 0.050, significant at 5 percent).

This regional deviation invites the same channel decomposition used for the OECD panel. Adding an ASEAN interaction to the wage, employment and value-added channels of Equation 6 (518-observation sample), the negative ASEAN deviation (-0.097, significant at 10 percent) runs almost entirely through the wage channel (-0.084) rather than employment (+0.015), the interaction channels satisfying the identity exactly $(-0.084 + 0.015 - 0.028 = -0.097)$. Whereas OECD industries raised employment enough to offset a wage discount, Vietnam's regional peers show the wage discount without a compensating employment gain, pointing to wages, not employment, as the margin along which labor's share erodes regionally. Leaving out each ASEAN country from this four-country panel (Indonesia, Malaysia, the Philippines, Thailand) in turn, the wage-channel coefficient stays negative in every fold (-0.046 to -0.119), so no single country drives the finding.

Table 8. ASEAN Spotlight, Parts A-C (Emerging-Market Panel)

	FE-IV EM (baseline)	FE EM + ASEAN interaction
Part A - AI Patent Intensity and Carbon Intensity		
AI Patent Intensity (ln, IV) [EM]	0.8010 (0.6950)	
AI Patent Intensity (ln) [EM]		0.0430 (0.0490)
AI Patent Intensity x ASEAN		0.0190 (0.1330)
Num.Obs.	591	591
R2 Within	0.0410	0.0300
Part B - Green Technology Intensity and Productivity		
Green Tech Intensity (ln, IV) [EM]	0.2130 (0.2030)	
Green Tech Intensity (ln) [EM]		-0.0130 (0.0320)
Green Tech Intensity x ASEAN		0.1980* (0.1100)
Num.Obs.	536	609
R2 Within	0.1220	0.0720
Part C - AI Patent Intensity and Labor Income Share		
AI Patent Intensity (ln, IV) [EM]	0.4550 (0.3390)	
AI Patent Intensity (ln) [EM]		0.0620* (0.0330)
AI Patent Intensity x ASEAN		-0.1020** (0.0500)

	FE-IV EM (baseline)	FE EM + ASEAN interaction
Num.Obs.	530	530
R2 Within	0.1130	0.1020

Table 9 reports a Vietnam-inclusive robustness check: replacing EPO AI-patent counts with WIPO counts, bringing Viet Nam into the sample, leaves the Part A coefficient indistinguishable from the EPO-based estimate (0.025, SE 0.030, versus 0.801), and for Part C gives a small, ASEAN-moderated coefficient (0.001, SE 0.019, with a significant ASEAN interaction of 0.056, significant at 5 percent) consistent with the main result. *WIPO's shorter, noisier series means this is reassurance, not a precise estimate for Vietnam - it shows the results aren't an artifact of Vietnam's absence from the main panel.*

Table 9. Vietnam-Inclusive Source Robustness: EPO vs WIPO AI Patents, Parts A and C (EM)

	FE-IV, EPO	FE-IV, WIPO	FE, WIPO	FE + ASEAN int., WIPO
Part A - AI Patent Intensity and Carbon Emission Intensity				
AI Patent Intensity (ln, IV) [EPO]	0.8010 (0.6950)			
AI Patent Intensity (ln, IV) [WIPO]		-0.3530 (0.9590)		
AI Patent Intensity (ln) [WIPO]			0.0250 (0.0300)	0.0330 (0.0340)
AI Patent Intensity [WIPO] x ASEAN				-0.0600 (0.0550)
GDP per capita (ln)	-0.5670 (0.6710)	-0.8040* (0.4170)	-0.7530* (0.3910)	-0.7490* (0.3920)
Renewable energy (%)	-0.0240 (0.0180)	-0.0280** (0.0130)	-0.0260** (0.0110)	-0.0260** (0.0110)
Num.Obs.	591	677	677	677
R2 Within	0.0410	0.0350	0.0350	0.0360
Part C - AI Patent Intensity and the Labor Income Share				
AI Patent Intensity (ln, IV) [EPO]	0.4550 (0.3390)			
AI Patent Intensity (ln, IV) [WIPO]		-0.4740 (0.8800)		
AI Patent Intensity (ln) [WIPO]			0.0090 (0.0180)	0.0010 (0.0190)
AI Patent Intensity [WIPO] x ASEAN				0.0560** (0.0270)
GDP per capita (ln)	0.0410 (0.3930)	-0.1110 (0.3550)	-0.0300 (0.2980)	-0.0340 (0.3000)
Trade openness (% GDP)	-0.0060** (0.0020)	-0.0030 (0.0040)	-0.0050*** (0.0020)	-0.0050*** (0.0020)

	FE-IV, EPO	FE-IV, WIPO	FE, WIPO	FE + ASEAN int., WIPO
Num.Obs.	530	616	616	616
R2 Within	0.1130	0.0810	0.0790	0.0810
FE: pair_id_em / year	Yes	Yes	Yes	Yes

Together, the evidence points one way: Vietnam and its ASEAN peers capture a stronger green-technology dividend than the wider EM group, but translate AI intensity into labor-share gains more weakly, or not at all - a genuinely two-sided finding, without reaching for a regional causal mechanism the data can't support.

6.4 Synthesis across parts

Figure 2b juxtaposes the three OECD FE-IV estimates and their EM counterparts. The OECD columns tell a consistent story: AI and green-technology patenting are associated with lower carbon emission intensity, higher labor productivity, and, once offsetting wage and employment channels are netted, no erosion of labor's share alongside a significant employment expansion. The EM columns all run toward a technology-dividend story, positive for labor share and productivity, with carbon intensity opposite the OECD sign but imprecise; only labor share reaches significance at $\alpha = 0.10$. This less precise EM pattern is at least as consistent with the EM panels' smaller cluster counts providing less power as with a different relationship: even in the far larger OECD panels the aggregate labor-share and productivity coefficients (0.0114 and 0.0187) fall below their minimum detectable effects at 90 percent power (0.0746 and 0.0203), so the smaller EM panels are a fortiori underpowered to distinguish such effects from zero, and the EM nulls bound rather than refute the OECD story.

Further robustness checks leave the conclusions unchanged: leave-one-country and leave-one-ASEAN-out re-estimation (no single country drives the results); a placebo test with randomly reassigned instruments (null coefficients); the Gollin (2002) self-employment adjustment (no change to the OECD decomposition); and a dynamic panel GMM estimator (Table 5), consistent in sign and magnitude with the main results.

7 Conclusion and Policy Recommendations

7.1 What the three research questions show

A single FE-IV framework, run across three margins in OECD and EM panels, answers the essay's three research questions. RQ1: AI patent intensity cuts OECD

carbon emission intensity (-0.0965, significant at 1 percent); the EM direction stays unclear at current precision. RQ2: green technology intensity lifts OECD labor productivity (0.0187, significant at 1 percent), more so above-median AI exposure and stronger still within ASEAN. RQ3: AI patent intensity shows no net effect on the OECD labor income share (0.0114, not significant) - but decomposed, that null hides a real employment gain (0.0899) nearly cancelled by a wage discount (-0.0706). The EM effect is modestly positive (0.0622, significant at 10%) and wage-driven, while ASEAN runs weaker or reversed.

7.2 Policy implications for Vietnam

Vietnam draws three lessons from this. AI adoption doesn't have to work against the 2050 net-zero target - if the OECD technique effect holds as local industries adopt AI-complementary methods, upgrading and emission cuts can move together, though the imprecise EM carbon estimate keeps that a hypothesis, not a conclusion. Green technology and AI adoption also reinforce each other's productivity gains in the OECD panel, more so within ASEAN, which argues for pairing AI-adoption support with green-patenting incentives rather than running them as separate tracks. Most actionable of all: ASEAN's labor-share gap runs through wages, not employment - a headcount metric would miss it entirely. Policy should track wage growth against AI-adoption intensity directly, not treat rising employment as proof workers are sharing in the gains. That's the middle-income-trap lesson: upgrading only counts if labor captures gains in wages, not just headcount.

7.3 Limitations and future research

Four limitations bear on these conclusions. Every EM estimate rests on a thin cluster count (67-75 pairs versus 75-250 in the OECD panels) - likely why EM coefficients come out less precise, though a non-result here doesn't prove no relationship exists. Mainland China and South Africa drop out of the labor-share panel because UNIDO INDSTAT4 lacks the needed value-added series, a real gap given China's weight in manufacturing. The wild-cluster-bootstrap fallback used for small-sample EM inference is a documented compromise, not the first-choice method. And both technology measures proxy patenting activity rather than adoption itself, likely undercounting diffusion where firms license or import technology instead of patenting it - which would bias EM coefficients toward zero.

Two extensions would sharpen these findings. Firm-level Vietnamese data - the General Statistics Office's enterprise survey, for instance - would test whether the industry-level patterns hold up at the firm level, and would substantially expand the effective sample beyond the seven-to-eight-country ASEAN panel used here. Matched employer-employee data on Vietnam's manufacturing wage-employment split would test directly whether the OECD pattern - employment gains offset by wage discounts - shows up in Vietnam too.

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